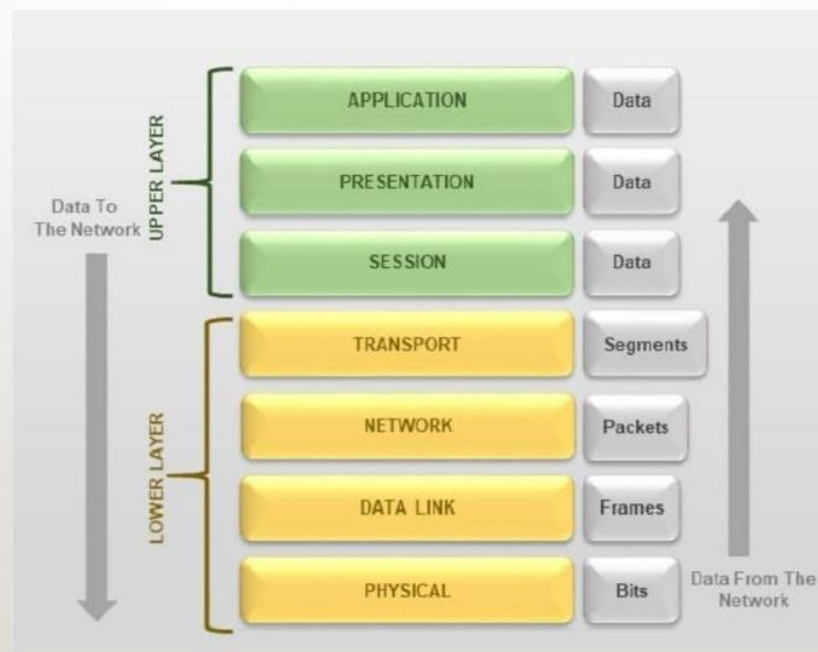


DIFFERENCE BETWEEN OSI MODEL AND TCP/IP MODEL

OSI MODEL	TCP/IP MODEL
Contains 7 Layers	Contains 4 Layers
Uses Strict Layering resulting in vertical layers.	Uses Loose Layering resulting in horizontal layers.
Supports both connectionless & connection-oriented communication in the Network layer, but only connection-oriented communication in Transport Layer	Supports only connectionless communication in the Network layer, but both connectionless & connection-oriented communication in Transport Layer
It distinguishes between Service, Interface and Protocol.	Does not clearly distinguish between Service, Interface and Protocol.
Protocols are better hidden and can be replaced relatively easily as technology changes (No transparency)	Protocols are not hidden and thus cannot be replaced easily. (Transparency) Replacing IP by a substantially different protocol would be virtually impossible
OSI reference model was devised before the corresponding protocols were designed.	The protocols came first and the model was a description of the existing protocols

OSI MODEL LAYERS



. Transmission Media

- **Guided Media:**
 - **Twisted Pair Cable:** Consists of pairs of wires twisted together. Categories include CAT5, CAT6. Used in LANs.
 - **Coaxial Cable:** Consists of a central conductor, insulating layer, metallic shield, and outer cover. Used for cable TV and broadband internet.
 - **Fiber-Optic Cable:** Uses light to transmit data. High-speed and long-distance communication. Types: Single-mode, Multi-mode.
- **Unguided Media:**
 - **Radio Waves:** Used for wireless communication like Wi-Fi, Bluetooth.
 - **Microwaves:** Used for long-distance wireless communication, satellite networks.
 - **Infrared:** Used in remote controls, short-range communication.

Network Topology

- **Bus Topology:** All devices share a single communication line. Simple and cost-effective but prone to collisions.
- **Star Topology:** All devices connected to a central hub/switch. Easy to manage and troubleshoot. If the central device fails, the network goes down.
- **Ring Topology:** Devices connected in a circular fashion. Data travels in one direction. A failure in one device can affect the whole network.
- **Mesh Topology:** Devices are interconnected. Provides high redundancy and reliability but is costly and complex.
- **Tree Topology:** Hierarchical structure combining star and bus topologies. Scalable and easy to manage.
- **Hybrid Topology:** Combination of different topologies to form a robust network.

Error Detection & Correction

- **Parity Check:** Adds a parity bit to make the number of 1s either even (even parity) or odd (odd parity).
- **Checksum:** Divides data into segments, sums them, and attaches the sum (checksum) to the data. Used in TCP/UDP.
- **Cyclic Redundancy Check (CRC):** Applies polynomial division on data. Used in network communications and storage devices.
- **Hamming Code:** Adds redundancy bits to data to detect and correct errors. Can correct single-bit errors and detect double-bit errors.

Media Access Protocols

- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):**
 - Used in Ethernet networks.
 - Devices check if the network is free before transmitting.
 - If collision occurs, devices wait for a random period before retransmitting.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance):

- Used in wireless networks (Wi-Fi).
- Devices signal their intent to transmit to avoid collisions.
- Uses acknowledgments to confirm successful transmissions.
-

Token Ring:

- Devices pass a token along the network.
- Only the device holding the token can transmit data.
- Reduces collisions but slower due to token passing

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VLAN (Virtual Local Area Network)

- **Functionality:**
 - Allows logical segmentation of a physical network.
 - Devices in a VLAN can communicate as if they are on the same physical network, regardless of their actual location.
 - Reduces broadcast traffic and improves security.

- **Implementation:**
 - VLANs are configured on network switches using the IEEE 802.1Q standard.
 - Each VLAN is identified by a unique VLAN ID.
- **Benefits:**
 - Improved network management and scalability.
 - Enhanced security by isolating sensitive data.
 - Simplified administration by grouping users by department or function.

Access Control List (ACL)

- **Purpose:**
 - Filters network traffic based on predefined rules.
 - Controls access to and from network segments.
- **Types:**
 - **Standard ACLs:** Filter traffic by source IP address. Limited flexibility.
 - **Extended ACLs:** Filter traffic by source and destination IP addresses, port numbers, and protocols. More granular control.
- **Usage:**
 - Applied to network interfaces on routers and firewalls.
 - Used to permit or deny traffic based on security policies.

IP Addressing

- **IPv4:**
 - 32-bit address divided into four octets.
 - Written in dotted decimal format (e.g., 192.168.1.1).
 - Limited address space, leading to IPv4 exhaustion.
- **IPv6:**
 - 128-bit address, written in hexadecimal format.
 - Provides a vastly larger address space.
 - Examples: 2001:0db8:85a3:0000:0000:8a2e:0370:7334.
- **Public vs. Private IP:**
 - **Public IP:** Globally unique, assigned by ISPs. Routable on the internet.
 - **Private IP:** Used within private networks. Not routable on the internet. Examples: 192.168.x.x, 10.x.x.x, 172.16.x.x to 172.31.x.x.

Routing

- **Static Routing:**
 - Manually configured routes.
 - Simple and secure but not scalable for large networks.
- **Dynamic Routing:**
 - Uses routing protocols to automatically adjust routes.
 - Examples: OSPF (Open Shortest Path First), BGP (Border Gateway Protocol), RIP (Routing Information Protocol).
 - Adapts to network changes and failures.
- **Routing Table:**
 - Database of network paths.
 - Contains information on destination networks, next-hop addresses, and metrics.

Subnetting

- **Purpose:**
 - Divides a large network into smaller, manageable sub-networks.
 - Enhances security and improves network performance.
- **Subnet Masks:**
 - Defines the network and host portions of an IP address.
 - Example: 255.255.255.0 for a Class C network.
- **CIDR Notation:**
 - Represents subnets with a suffix indicating the number of network bits.
 - Example: 192.168.1.0/24.
- **Calculating Subnets:**
 - Determine the number of required subnets and hosts per subnet.
 - Adjust the subnet mask accordingly to provide the necessary number of subnets and hosts.
- **NAT (Network Address Translation):**
 - Maps private IP addresses to a public IP address.
 - Conserves public IP addresses.
 - Types: Static NAT, Dynamic NAT, PAT (Port Address Translation).
- **ARP (Address Resolution Protocol):**

- Resolves IP addresses to MAC addresses.
- ARP Request: Broadcast to find the MAC address for a given IP.
- ARP Reply: The device with the matching IP responds with its MAC address.

- **DHCP (Dynamic Host Configuration Protocol):**
 - Automatically assigns IP addresses to devices.
 - Provides IP address, subnet mask, gateway, and DNS servers.
 - Reduces manual configuration errors and simplifies network management.

Sliding Window Protocol

- **Function:**
 - Manages the flow of data packets between two devices.
 - Ensures reliable and sequential delivery of frames.
 - Uses a window to control the number of frames sent before needing an acknowledgment.
- **Types:**
 - **Go-Back-N:** Sender can send several frames before needing an acknowledgment but must resend all frames after a lost frame.
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Collision Domain and Broadcast Domain

- **Collision Domain:**
 - A network segment where data packets can collide.
 - Collisions occur in shared media environments like Ethernet hubs.
 - Reducing collision domains improves network performance.
 - Switches and routers can segment collision domains.
- **Broadcast Domain:**
 - A network segment where a broadcast frame is forwarded to all devices.